Guidance and Control of Unmanned Autonomous Vehicles

(Course Code: AE 700)

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Practice Set – 1

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1. Assume that the orientations of two coordinate frames are related using Euler angles, where $ZY'X''$ sequence is considered. Find out the necessary Euler angles (ψ, θ, ϕ) from the transformation matrix given below, wherein some of the entries are missing.

$$T = \begin{bmatrix} a & -0.7189 & b \\ 0.7166 & -0.6732 & 0.1824 \\ -0.0872 & 0.1730 & c \end{bmatrix}$$

(**Ans:** $a = -0.69203$, $b = 0.06537$, $c = 0.98109$)

2. Consider a scenario where the accelerometers are mounted on the aircraft in strap-down mode, which essentially means that the measurements are taken in the body frame. If the orientation of aircraft body axes, at some time instant, are specified by sequential rotations using Euler angles of $60°, 45°$, and $35°$ in roll, pitch, and yaw angles $(XY'Z'')$, respectively. Suppose the accelerometer outputs at the same time instant are measured as $(5, 4, 3)^\top$ m/s^2. What would be the components of aircraft acceleration in the inertial frame at that time instant?

Ans:$(3.395, 2.338, 5.745)^\top$ m/s^2

3. Determine the Euler angles associated with the rotation matrix $(ZY'X'')$ given below. Also, explicitly specify the quadrant to which each Euler angle belongs.

$$R = \begin{bmatrix} -0.1422 & 0.0996 & 0.9848 \\ -0.6949 & 0.6985 & -0.1710 \\ -0.7049 & -0.7086 & -0.0302 \end{bmatrix}$$

(**Ans:** $\phi = -100°$, $\theta = -80°$, $\psi = 145°$)

4. Find the transformation matrix for the given proper Euler angles. The angles corresponding to three intrinsic rotations are $30°$ (about Z-axis) , $45°$ (about Y'-axis), and $60°$.

Ans:

$$\begin{bmatrix} -0.1268 & 0.9268 & -0.3536 \\ -0.7803 & 0.1268 & 0.6124 \\ 0.6124 & 0.3536 & 0.7071 \end{bmatrix}$$

5. Consider the scenario where one coordinate frame is rotating with respect to other fixed coordinate frame with an angular rate of $\boldsymbol{\omega}$. If the unit vectors of rotating and fixed coordinate frames are denoted by $(\boldsymbol{i}, \boldsymbol{j}, \boldsymbol{k})$ and $(\boldsymbol{I}, \boldsymbol{J}, \boldsymbol{K})$, respectively. Prove that the rate of change of *unit vector of rotating coordinate frame* with respect to *fixed coordinate frame* can be obtained by $\dfrac{d\boldsymbol{i}}{dt} = \boldsymbol{\omega} \times \boldsymbol{i}$.

6. Find out the Euler angle rates $(\dot{\theta}, \dot{\phi}, \dot{\psi})$ in terms of body rates (p, q, r) for the case where the orientations of two frames are related using Euler angles (θ, ϕ, ψ) in $YZ'X''$ sequence. Obtain the points of singularities for this sequence. Also, find out the above relation in the vicinity of any one point of singularity if there are multiple such points.

   Hint: You may use the limiting value approach to obtain such a relation.

7. Suppose that the orientation of a frame attached to a rigid-body aircraft with respect to an inertial frame is specified using Euler angles of $30°, 45°, 60°$, with $Z - Y' - X''$ sequence of rotations. Suppose the designer aims to use a quaternion to specify the same orientation, as specified above by Euler angles. What would be the necessary value of quaternion to specify the same orientation of aircraft? Consider all Euler angle rotations to be intrinsic in nature. Also, find the equivalent quaternion when the sequence of Euler rotations is reversed $(X - Y' - Z'')$ while keeping angles the same

   (**Ans:** a) $q_o = 0.8224$, $q_1 = 0.3604$, $q_2 = 0.4397$, $q_3 = 0.0223$
   b) $q_o = 0.7233$, $q_1 = 0.5320$, $q_2 = 0.2006$, $q_3 = 0.3919$)

8. Consider a rotation of a vector, using quaternion, about an axis defined by the vector $(1, 0, 1)$ through an angle of $2\pi/3$.

   (a.) Obtain the quaternion $\mathcal{Q}$ to perform this rotation.
   (**Ans:** $Q = 0.5 + 0.6124\hat{i} + 0.6124\hat{k}$)

   (b.) Compute the effect of rotation on the basis vector $\mathbf{j} = (0, 1, 0)$.
   (**Ans:** $v = -0.6124\hat{i} - 0.5\hat{j} + 0.6124\hat{k}$)

   (c.) Find the coordinates of the above vector in the new frame if we rotate the coordinate frame itself about the same axis and angle while keeping the vector constant?
   (**Ans:** $v = +0.6124\hat{i} - 0.5\hat{j} - 0.6124\hat{k}$)

9. If an airplane is rotated, first along the Z axis with an angle ψ in the counter-clockwise direction, then along the rotated Y axis with an angle θ in the clockwise sense, followed by a counter-clockwise rotation along newly formed X axis by an angle ϕ.

   (a.) Compute the Euler angle transformation matrix for composite rotation.

   (b.) If the above composite rotation is represented by using a single quaternion, say $\mathcal{Q} = q_0 + q_1 i + q_2 j + q_3 k$, then find this quaternion in terms of the Euler angles.

10. If the orientation of the body frame with respect to the earth-fixed frame is specified using three sequential rotations about Z,Y, and X with angles ψ, θ, and ψ, respectively. (The aircraft is assumed to be rigid, and the earth is assumed to be flat and stationary in inertial space). The Euler rates are assumed to be given by $\dot{\psi} = 2$ rad/sec, $\dot{\theta} = 3$ rad/sec, and $\dot{\phi} = 5$ rad/sec. Then, compute the values of body rates for $\psi = 30°$, $\theta = 45°$, and $\phi = 60°$. Furthermore, Assume that the aircraft maintains the same body rates for the next 2 secs. Then, find the Euler rates after 2 secs.

11. Suppose, the aircraft is headed toward your favorite destination of your choice. Find out the course angle of that destination(or maybe assume 37°). It is given that the flight path angle is 45° and the aircraft is moving with the speed of 230m/s with respect to ground. There is constant wind of 10m/s in the east direction. The Euler angles of aircraft with respect to the NED frame are $30°, 45°, 60°$, with $Z - Y' - X''$ sequence of rotations. Find out the angle of

attack, side-slip angle, and airspeed (α, β, V_a)
Hint: Using course angle, flight path angle, and ground speed compute V_g in NED Frame. Similarly compute wind velocity V_w in NED frame. Then get the aircraft velocity w.r.t in wind in NED frame. Now, with the help of Euler angles and rotation matrix get the components in the body frame. Now, (α, β, V_a) can easily be computed. **Ans:** $(\alpha = 88.03°, \beta = 79.54°, V_a = 223.62\text{m/s})$

12. Determine the relation between the body angular velocity vector and the Euler angle rates in the sequence $2 - 1 - 2$.

13. The initial $3 - 2 - 1$ Euler angles yaw, pitch, and roll of a vehicle are $(\psi_0, \theta_0, \phi_0) = (40, 30, 80)$ deg at time t_0. Assume that the angular velocity vector of the craft is given in body frame components as

$$\omega = \begin{pmatrix} \sin(0.1t) \\ 0.01 \\ \cos(0.1t) \end{pmatrix} 20 \text{ deg/s}$$

The time t is assumed to be given here in units of seconds.

(a) Translate this initial attitude description into the corresponding Euler parameters.

(b) Write a program to numerically integrate the Euler parameters over a simulation time of 1 min. Note that you must integrate these equations using radians as the angular units. Plot the four Euler parameter time histories.

(c) Given the result of the numerical integration, plot the Euler parameter constraint $\beta_0^2 + \beta_1^2 + \beta_2^2 + \beta_3^2$; comment on these values.

14. Let the $\mathcal{B}$ frame attitude relative to the $\mathcal{N}$ frame be given by the (3-2-1) Euler angles $(10, 25, -15)$ deg. Find the corresponding principal rotation axis and angles.

Ans: principal rotation axis $\hat{\mathbf{e}} = \begin{pmatrix} -0.532035 \\ 0.740302 \\ 0.410964 \end{pmatrix}$ and the first principal rotation angle $\Phi = 31.7762$ deg and the second $\Phi' = -328.2238$ deg

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